DARK CRIMENET, An Offline Investigation-Intelligence Platform for FIR Analysis, Link Prediction and Tamper-Evident Case Records

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Abstract: Police Officers get First Information Reports (FIRs), sometimes which may be scanned, containing unstructured text in different languages. This unstructured text contains entities, phone numbers and narratives that can be correlated between cases that are difficult to understand just by reading them. In this paper we provide DARK CRIMENET which reads through the text and/or scans of FIRs, extracts entities, constructs a graph per case, suggests missing links, and maintains a tamper-proof audit trail. Missing links are found with the help of a self-supervised logistic regression model. We use weighted scores to rank entities Every action we performed in the application is logged in a secured signed hashed ledger. The model does not use any deep learning models and can be operated entirely offline if needed. We were able to achieve an accuracy of 99.6% using the context classifier. Also, with the help of link predictor, we were able to achieve AUC score of 0.85-0.96 compared to the AUC score of 0.24-0.29 of common neighbor method. Note: We used synthetic data for training and testing purpose. These are therefore upper bounds.

Index Terms: FIR analysis, investigation intelligence, link prediction, entity extraction, graph analytics, tamper-proof audit.

1. Introduction

A First Information Report is generally where an Indian criminal study begins, so even though, it's written as a narrative, the names, phone numbers, bank accounts, vehicle numbers, places and dates sit inside sentences, sometimes typed, sometimes scanned from paper, and in English, Hindi or Marathi, in real-world scenarios. An investigator who wants to know whether the phone in one complaint also appears in another must read both and remember, and as, the number of cases is currently expanding, links between people and accounts that are normally present in the entries stay unseen.

Existing work on FIRs has mainly focused on the filing step, like letting a citizen summarize an incident, turning speech to text, and recommending IPC sections [1]; these frameworks are only helping in producing a correct FIR. Far fewer tools help after the FIR exists, when the questions are who is connected to whom, which connection is probably missing, and whom to look at first; another big concern is trust, investigation records must show that nobody has changed them quietly. Also, the tools should be able to run on ordinary police computers that may (to some degree) have no internet access and no GPU, a graphics processor.

This paper presents DARK CRIMENET, a system that is built around those needs. We can upload any FIR document. It will then scan it, translate it, if necessary, extract important information from the document, create graphs, find missing links, highlights possible suspects, scores them using an explainable score and logs all activity into a signed ledger using hashing. All decisions can be tracked back to a rule, weighting or mini model so that the system can be used within an investigation where the officer will need to be able to explain why something has been shown. This is why deep learning models were avoided.

The progress made by this project so far includes:

1) Offline data extraction system which uses a labelled parser alongside checksum checkers and a context classifier.
2) Self-supervised linking algorithm with no external training data as it uses the case graph itself for training.
3) Scoring of possible suspects based on scores with explanation.
4) Tamper-proof ledger using Ed25519 signatures, witness files and TOTP two factor login (for now).
5) Comparison of model and clear explanation of why it's or isn't being not used

The rest of the paper is mainly organized as follows; Section II reviews related work, Section III

presents an overview of the framework, Section IV describes the data and set-up, Section V detail the findings and model comparisons, Section VI tries out the results, Section VII is covering shortcomings and future work, Section VIII is the concluding section.

2. Related Work

Work related to this paper falls into the three groups - FIR drafting and legal-section prediction, legal language models, and network analysis of crime data.

FIR drafting and section prediction: An automated system was designed by Mathivanan et al. [1] that allows text/audio to be inputted, and be able to extract information regarding the victim, suspect, date and location and also suggests relevant IPC sections with their confidence levels. all the Evidence can be stored securely using a blockchain inspired system. 2500 FIRs and 500 audio samples can be reported on. A fine-tuned LegalBERT model is used to predict the sections, with an accuracy of 84%. This has helped in significantly reducing the amount of time needed to draft an FIR from ~25mins to less than 5. This survey also talks about systems like the Smart IPC Section Prediction. These use light-weight transformer models e.g. DistilBERT, TinyBERT and JurisAI. It also allows audio input and template-based drafting. The abstract of [1] describes that there is an average accuracy of 93% however the results section and Figure 4 show the accuracy to be lower at 84%. Overall, this group of systems generally focus on assisting in FIR drafting rather than analysing the content within the FIRs.

Legal models: The legal domain already contains many transformer models that are trained on legal texts (e.g., LEGAL-BERT [13]). This results in a legal bias in these large models during application. However, [1] makes use of this finding. These models are most of the times expensive because of the data required and hardware that is needed for fine-tuning and serving these models on FIR tasks. Real FIR data is also confidential, so most of the time it's difficult to find this kind of data for a student or a small project. Therefore, we decided to train our models on synthetic data and only give an upper bound on how accurate they will be.

Link prediction and graph analysis: Link prediction is generally asking which pairs of nodes that aren't yet connected are likely to be connected [5], the classic things, such as common neighbours, scoring a pair by the contacts it's sharing. They're working well in social networks where friends of friends become friends. PageRank [9] is broadly used, and it works when the network is well gathered for ranking important nodes, but still, in a criminal network the missing link is often between two people who are sharing no contact at all.

TABLE I
POSITIONING AGAINST THE SYSTEMS REVIEWED IN [1]

System	Method	Limitation
Reference system [1]	LegalBERT, NER, ASR	Drafting-focused; GPU; no network analysis
Smart IPC Section Prediction (in [1])	DistilBERT, TinyBERT	Section prediction only
JurisAI (in [1])	NLP, speech, templates	Drafting only
This work	Parser, regex, LR, graph	Synthetic data only so far

Overview: The work described here is shown in Table I. To the best of our knowledge, we didn't find any system described in [1] that have all three features described (link prediction between cases, explainable lead ordering, signed audit log) in an offline package. We are mostly focusing only on those systems that are described in [1]. A more general literature review will be conducted as future work.

3. Proposed System

3.1. Overview

In Fig. 1 we can see the overall pipeline of the project. The record enters as text, PDF or image, and is passed through OCR and language identification, then translated to English if needed, and is then scanned for items, from all the elements extracted a graph is then formed, and the graph was read by four analysis modules, their outputs are also shown in a dashboard with the help of a map, a timeline, we also implemented a rule-based only question-answering copilot and reports.

3.2. Ingestion and translation

Scanned pages are read with Tesseract 5 [12] using its LSTM engine, page-segmentation and several preprocessing variants, with language packs for English, Hindi and Marathi (more languages can be added later); the language of each sentence is identified from its script and function words. Translation is mostly rule-based and fully local. Sensitive Information or Identifiers such as phone numbers and account numbers are protected so they're passing through without getting changed, words are translated with a glossary and reordered, names are transliterated, places are matched against a record,

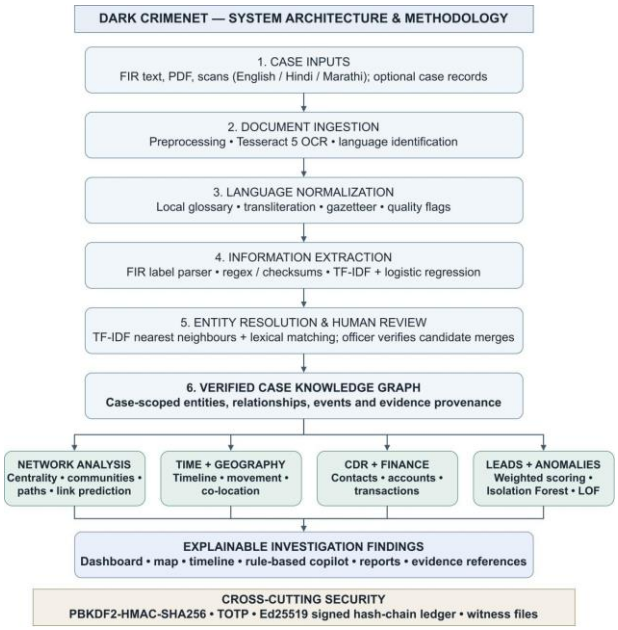


Fig. 1. System architecture of DARK CRIMENET (made in draw.io).

and an optional neural translator is used only if it's installed. Also, every sentence receives a quality flag, like words that can't be translated are kept in double brackets so that a reader can identify those easily and see what was not understood.

3.3. Item extraction

Labels like "Complainant Name:" or "Mobile:" are, generally present in the FIRs. A label-driven parser finds those and reads them first. Free text is then scanned with regular expressions for information like phone numbers, accounts, vehicle numbers, PAN and IFSC patterns, with checksum validators (to make the Verhoeff check for Aadhaar-style numbers) to reject strings that only look right in the application. Another check like, a context classifier goes through the words that are pulled from the scan to see if they fit the wanted field type; it classes person, phone, account, vehicle, location, case, date, time, organization, and is using TF-IDF and logistic regression, that are trained on 2,700 templated snippets. We map section numbers to Act through a lookup, which has 0.984 precision on our gold set. Also, the model file is hashed before its normally stored and the loader will refuse a file whose hash doesn't match, because such files can (in practice) codify when loaded.

3.4. Case graph and entity resolution

The same person may be written differently in separate FIRs, we tried to solve this issue by using TF-IDF nearest neighbors combined with string similarity (0.68 cosine and 0.32 lexical weight, merge threshold 0.72), the elements will become nodes and recorded relations will become edges; merges are not applied silently; they are put forward first for the officer to verify them.

3.5. Link prediction

The model is calculating 12 different features from each item pair. For example, common neighbors, graph distance, spectral similarity, how often entities connect and case/event overlap in the data. We have implemented a self-supervised model and for every round we are keeping 20 percent of the links hidden for its training. Also, we are using the negative sampling setting which gives three times as many non- links as positives to train our model better. The model also learns from the graph structure. We were able to do all this with the help of logistic regression, with gradient boosting as the baseline reference. From all this we were able to produce ranking signal rather than some adjusted probability; We also make sure that our interface shows why that recommendation/link is recommended by showing what was the reason it was selected for.

3.6. Lead scoring and anomaly detection

We are able to calculate the lead-priority score by the help of differently weighted behavioral and graph signals, we chose this so that each advancement is visible to the investigator and isn't just trained centrally; we have also tried to mention it wherever we could in our project. Anomaly detection is possible because of Isolation Forest and Local outlier factor on per entity credits; we also show which feature contributed the most for getting it flagged.

3.7. Security and audit

Passwords are hashed with the help of PBKDF2-HMAC-SHA256 with 210,000 iterations, each session is assigned tokens that are re-checked against the database; and so, to change a record unnoticed, an attacker would need the signing key, the witness file and every later block. Also, the same for deleting, the end of the chain is detectable, really, a Merkle root is anchored every 50 blocks and a separate witness file records chain heads, in the application. We also implemented two step sign-in which is following

TABLE II
MODULE SUMMARY

Module	Method	Trained	Measured
Entity context	TF-IDF + LR	Yes	Yes
FIR parser, validators	Rules, checksums	No	Yes (30 FIRs)
Translation	Glossary, rules	No	Quality flags
OCR	Tesseract 5	Pre-trained	No
Entity resolution	TF-IDF KNN + string	No	No
Link prediction	LR, 12 attributes	Self-supervised	Yes
Anomaly	Isolation Forest + LOF	Unsupervised	No
Lead score	Weighted blend	No	Yes (imitated)
Social flags	97 rules	No	Yes
Copilot	Intent matching	No	—

TABLE III
DATA USED FOR EACH EVALUATION

Module	Data	Size	Split
Entity context	Templated FIR snippets, 10 classes	2,700	80/20 hold-out + 5-fold CV
Extraction	Hand-labelled FIRs	30	18 dev, 12 held-out
Link prediction	Demo cases, own graph	3 cases	Hidden 20% links, 3 repeats
Suspect ranking	Behaviour simulator	3 levels	Known ground truth
Social flags	Labelled posts, 97 rules in 7 categories	—	Dev + held-out

the TOTP standard; a 30-second, six-digit encrypted code is generated along with ten hashed recovery codes and a locally created QR code; every action is added to a ledger in which each block is holding the hash of the previous block and an Ed25519 signature [11].

3.8. Map, timeline and copilot

To understand when and where each event happened better, we tried to place every event on a map in order of reliability; more precisely, coordinates recorded on the event are used first, then coordinates of a linked location entity, and only then an offline gazetteer lookup of the place text, which is marked approximate, because its exactness is about 5-10 km; we also make sure that these frameworks stored on people, phones or accounts are never used, so a suspect can't be pinned to a wrong place by accident. The copilot answers questions by matching the question to a fixed set of intents, such as "who is connected to this number", and returns the answer with the data it came from and a disclaimer; it doesn't create free text, so it can't invent a fact that isn't in the case.

TABLE IV
ENTITY-CONTEXT MODELS ON THE SAME SPLIT (540 TEST SNIPPETS)

Model	Hold-out acc.	Macro-F1	5-fold CV	Value-only acc.
Char TF-IDF + Logistic Regression (used)	99.63	99.60	99.44 ± 0.12	90.56
Char TF-IDF + Linear SVM	99.63	99.60	99.44 ± 0.12	90.56
Char TF-IDF + Random Forest (200)	99.63	99.60	99.67 ± 0.14	83.33
Word TF-IDF + Logistic Regression	99.44	99.40	99.37 ± 0.25	86.30
Char TF-IDF + Multinomial NB	96.48	96.08	94.89 ± 1.17	87.04
Char TF-IDF + k-NN (k=5)	95.74	95.40	94.00 ± 0.71	82.78

3.9. Implementation

The back end is written in Python with FastAPI and SQLAlchemy over SQLite, the graph code uses NetworkX, and the models use scikit-learn; Table II summarizes each module, the method behind it, whether it's trained, and whether its precision has been measured, to its most. The front end is made using a single-page application in plain JavaScript with Cytoscape for graphs and Leaflet for maps, online street tiles are an opt-in layer. Both libraries are bundled with the application, and the base map is an India outline with a latitude and longitude grid, so the interface is working with no internet connection, and the whole framework is, as it stands, starting from one script on an ordinary Windows machine and needs no GPU, broadly speaking.

4. Datasets and Experimental Set-up

We were not able to train our models on real FIRs because those are confidential, so every dataset used here is synthetic; in fact, Table III is listing them. Neither the data nor the splits were tuned to favor a model; python is being used with scikit by all experiments-learn [3] and NetworkX [4], a fixed random seed (26189) and no GPU, broadly speaking.

Metrics are mainly precision and macro-F1 for grouping, ROC-AUC and Precision@k for link prediction, and AUC for lead ranking; precision@k is the share of the top k suggestions that are generally hidden real links; a random ranking scores 0.25, because of the 1:3 positive-to-negative ratio.

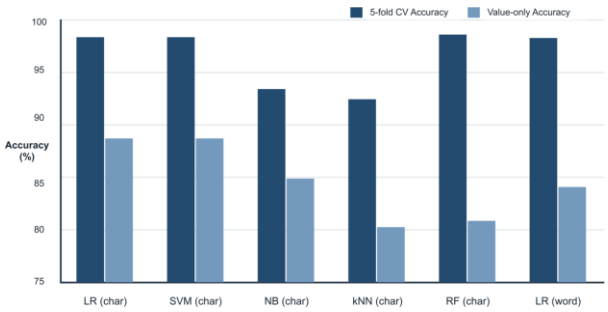


Fig. 2. Cross-validated and value-only exactness of the six models.

5. Results and Model Comparison

5.1. Entity-context classifier

To see which model is best for our application we tested six models on the same data split. In Table IV we can see that four out of the six models performed similar in terms of accuracy only. This tells us that we cannot just use precision when choosing our models. So, we tried a different approach of testing the models by giving only the values and hiding the labels, basically a stress test to see which model performs better even if their accuracy is the same. For example, we would only keep Rahul Sharma instead of Complainant Name: Rahul Sharma. The highest drop was in Random Forest from 99.63 percent to 83.33 percent only, highest was maintained by both logistic regression and linear SVM: almost 90 percent. The final choice was Logistic Regression as it had the same highest exactness, had the most consistency with missing labels and gave class probabilities which the SVM does not. The data here is templated therefore they will be an upper bound and do not represent the accuracy of the model on real FIRs.

5.2. Link prediction

The AUC values for the three example cases are provided in Table V. The Precision@k values are presented in Table VI. The logistic regression model performs well in general achieving between 0.85 and 0.96. The common neighbors and Adamic-Adar models do not perform well achieving between 0.24 and 0.29. This is not due to any issues with the baselines. Since the hidden links in these examples often connect entities that do not have any common contacts they will be given the lowest scores by the heuristics. However, the feature-based model does consider distance using the similarity and entity type pattern features. Gradient boosting does not generally perform better than the logistic regression model. It is also much less stable on the small graphs (with Case 101 having a standard deviation of 0.143). Most of the graphs are small with only 8 hidden links in Case 101 and 5 hidden links in the rest of the cases. For now, therefore, this shows trends, rather than accuracy.

TABLE V
LINK-PREDICTION AUC (MEAN ± SD, 3 REPEATS)

Method	Case 101	Case 103	Case 108
Logistic Regression (employed)	0.854 ± 0.021	0.962 ± 0.027	0.929 ± 0.049
Gradient Boosting	0.820 ± 0.143	0.964 ± 0.030	0.891 ± 0.069
Common neighbours	0.292 ± 0.039	0.244 ± 0.016	0.256 ± 0.016
Adamic-Adar	0.281 ± 0.060	0.244 ± 0.016	0.256 ± 0.016
Random guess	0.500	0.500	0.500

TABLE VI
PRECISION AT K (K = 8, 5, 5; RANDOM = 0.25)

Method	Case 101	Case 103	Case 108
Logistic Regression	0.58	0.87	0.67
Gradient Boosting	0.54	0.80	0.67
Common neighbours	0.13	0.00	0.00
Adamic-Adar	0.04	0.00	0.00

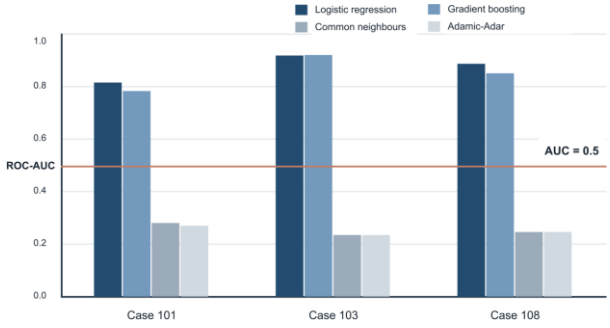


Fig. 3. Link-prediction AUC on the three demonstration cases.

TABLE VII
SUSPECT-RANKING AUC ON SIMULATED NETWORKS

Method	Easy	Standard	Hard
Hand-weighted, behaviour only	0.98	0.92	0.64
Hand-weighted + partial functions	0.99	0.96	0.79
PageRank	0.999	0.908	0.421

5.3. Lead ranking

For artificial networks where the true values are known, the weighted score performs very well, achieving AUC values of 0.98, 0.92 and 0.64 for the easy, standard and mainly challenging networks respectively when only behavioral information is used. For networks where some node role information was available it achieved even better performance of

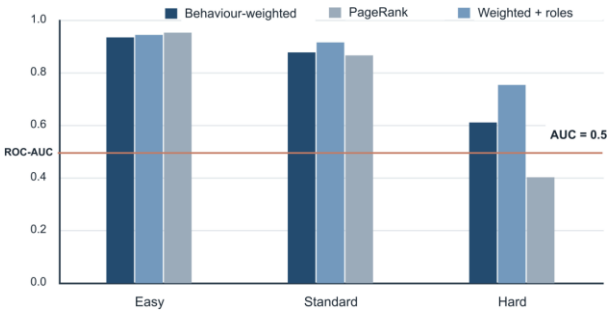


Fig. 4. Lead-ranking AUC on simulated networks of increasing hurdle.

TABLE VIII
FEATURE COMPARISON WITH [1]

Feature	[1]	This work
Voice complaint input	Yes (ASR)	No
Section recommendation	LegalBERT	Rule-based lookup
Entity extraction	BERT NER	Parser + regex + LR
Network / link analysis	No	Yes
Lead ranking	No	Yes (explainable)
Runs offline, no GPU	Not stated	Yes
Tamper-evident log	Blockchain-inspired	Ed25519 hash chain
Two-step sign-in	MFA (stated)	TOTP, RFC 6238

0.99, 0.96 and 0.79 (see Table VII). PageRank outperformed our method in the easy network (AUC = 0.999), however performed significantly worse than random guessing with a score of 0.42 for the challenging network. This was mainly because we intentionally removed the key individuals. This shows us that a metric that is only structural will most probably fail where there is an incomplete structure

5.4. Other modules and comparison with [1]

During the testing of models on 30 manually labelled FIRs (60/40 split), we managed to achieve a score of 1.00 F1 for information extraction tasks. Additionally, we achieved 0.984 precision accuracy for mapping of sections to Acts. Our current calibration error is 0.074 (in our environment). However, since the data and tasks are different, we cannot directly compare them. [1] focuses on drafting the FIRs, whereas our work focuses on after the drafting of the FIRs. We also compare the features of [1] against our work which you can see in Table VIII. We achieved an overall precision of 0.97, recall of 0.89 and F1 score of 0.92 for social media where we have almost 97 rules. We achieved high precision of 1.00 however recall of 0.71 gave us a F1 score of 0.83 for our test posts. We also managed to achieve a low false positive rate of 2.5%.

6. Discussion

From these results we can make some observations, that secondary factors played a major role in making most of the decision. Some of the things we wanted to make sure our work has been good performance even when the labels are missing, probability-based outputs, and quick execution times irrespective of the computational power. We also determined that using just purely structural metric for comparison won't work where most of the things are unstructured like in this case. In general, there was not a whole lot of difference between the top four entity classifiers (all within a fraction of a point). We also found that how we evaluated our models was more important than which models we are using. If we were to only look at neighbor heuristics, then link prediction seems to be the worst. However, if we decided to evaluate our predictions against a document that is hiding true links then we see it performing best. In fact, often those in the center were the least visible. For this type of data, we think the job can easily be done by simple models.

Threats to validity. The use of hidden known links means that the construct validity is concerned more with retrieving known links rather than discovering truly unknown links. All figures include these limitations meaning that they are indeed reported and any claims to the real world are avoided. In terms of external validity, small example graphs or simulated networks do not reflect all the different variety of real cases. In terms of internal validity, the entity snippets are template-based meaning that near duplicates can occur between the test and training sets. We also tried to make sure that no test text was used during training to best of our attempts.

Responsible usage: We ensure that all suggestions flagged as leads for a human officer will have the final decision made by a human. We aim to show the rules behind because of which each suggestion was made. To make sure rankings were fair, we used probabilities as a ranking indicator. The access can also be limited by enabling two steps to sign in. All actions are logged in the signed ledger to ensure misuse of the framework can be identified (to a significant extent). The whole pipeline has been covered by 352 automated tests.

7. Shortcomings and Future Work

Our major issue is that we don't have access to any real-world data/FIRs. The link prediction graphs for example are currently very small. The pipeline will

be evaluated using new work on public FIR/legal text. However, a sample will be taken to evaluate how well the data is extracted. Comparison of a transformer-based NER model will also be performed against our pipeline. Voice input might also be added to the system in future. If labelled outputs are available, the weights for lead score will be learned. In many ways the lead scores have been manually weighted. The OCR accuracy, entity resolution accuracy and anomaly detection accuracy haven't been measured. The entity classification has been trained on template-based snippets meaning there may be some near duplicates in terms of architectures. Every dataset is synthetic/invented meaning that the numbers given are upper bounds. The system is a decision support tool for a human officer meaning there will never be any guilty findings.

8. Conclusion

We have developed an offline solution DARK CRIMENET to transform FIRs into case graphs, suggest missing edges, rank investigative leads, and provides a tamper proof, digitally signed system. The classical models were sufficient for the tasks required for measurement. We applied a character level logistic regression model to provide context to entities, a feature based logistic regression model to predict edges, which clearly outperformed neighbor-based heuristics, and an explainable weighted scoring system to rank leads, which performed better than PageRank in the case where the network was hidden. But we were only able to use generated/synthetic data during testing to get results by our models. Further work will be to test this on real FIRs.

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